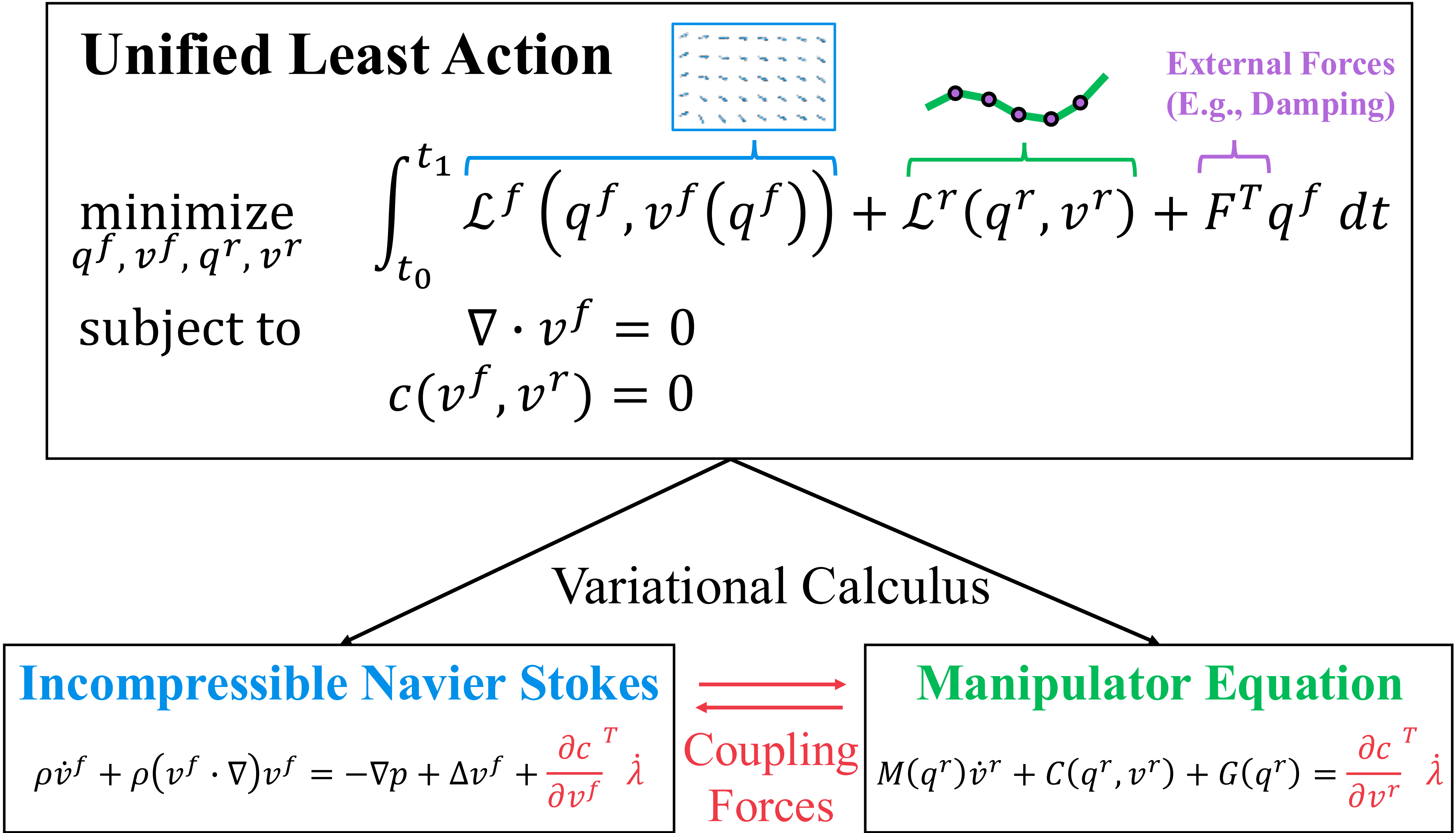
Unified Least Action
External Forces
(E.g., Damping)
$$\underset{q^f, v^f, q^r, v^r}{\text{minimize}} \int_{t_0}^{t_1} \mathcal{L}^f\left(q^f, v^f(q^f)\right) + \mathcal{L}^r(q^r, v^r) + F^T q^f \, dt$$
$$\text{subject to} \quad \nabla \cdot v^f = 0$$
$$c(v^f, v^r) = 0$$
Variational Calculus
Incompressible Navier Stokes
$$\rho \dot{v}^f + \rho(v^f \cdot \nabla)v^f = -\nabla p + \Delta v^f + \frac{\partial c}{\partial v^f}^T \dot{\lambda}$$
Coupling
Forces
Manipulator Equation
$$M(q^r)\dot{v}^r + C(q^r, v^r) + G(q^r) = \frac{\partial c}{\partial v^r}^T \dot{\lambda}$$